

Let's keep practicing real data analysis with the data sets we have seen: look at, plot, and analyze the data as required below. Please describe the data, your analysis, and your findings using complete sentences. **NOTE: While you may use other functions to check your work, you MUST show your calculations in this assignment.**

1. Using the dataset `carnivora` within the `ape` package, focus on the **gestation length** variable and compare the two SuperFamily samples, Caniformia and Feliformia. You can refer to the **Homework 08 Helper.Rmd** file for guidance on separating the groups. Include the following steps:
 - (a) Provide a short summary (2-3 sentences) of the dataset and the variable you are analyzing. After loading the data, you can use the command `?carnivora` to learn more about this data set, including which variable refers to gestation length.
 - (b) Construct at least one appropriate plot for visualizing the data. You may want to consider a separate plot for each SuperFamily.
 - (c) Define the variables describing the data – sample statistic, sample size, and sample error. These will be important for finding your confidence interval and performing your hypothesis test below.
 - (d) Perform a two sample hypothesis test for the means of the two SuperFamilies in terms of **gestation length**, with clear definitions for both hypotheses. List your test statistic and p-value, and use $\alpha = 0.05$ for your level of confidence.
 - (e) Construct an appropriately matching confidence interval (use the same α) and give your interpretation (e.g. “We are X% confident...”).
 - (f) List the statistical conclusion of your results (reject/fail to reject the null hypothesis).
 - (g) List a conclusion that you can explain to a non-statistician (what does your statistical conclusion say about the variable in question?).
2. Using the dataset `finches`, focusing on the **upperbeaklength** and **lowerbeaklength** variables, include the following:
 - (a) Provide a short summary (2-3 sentences) of the dataset and the variables you are analyzing. Note that these measurements are in millimeters.
 - (b) Construct at least one appropriate plot for visualizing the data.
 - (c) Define the variables describing the data (as in problem 1).
 - (d) Perform a paired hypothesis test for whether there is a difference of **8 mm** between **upper** and **lower beak length**, with clear definitions for both hypotheses. List your test statistic and p-value, and use $\alpha = 0.01$ for your level of confidence. Note that this is an uncommon case where the null hypothesis value is NOT 0.
 - (e) Construct an appropriately matching confidence interval and its interpretation.
 - (f) List the statistical conclusion of your results.
 - (g) List a conclusion that you can explain to a non-statistician.